

Chanukya_Walmart_Case

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1 Walmart Black Friday Analysis

1.1 Introduction

Walmart Inc., founded by Sam Walton in 1962 and headquartered in Bentonville, Arkansas, is a leading global retail corporation that operates thousands of supercenters, discount department stores, and neighborhood markets across multiple countries. Combining a massive brick-and-mortar footprint with a growing e-commerce platform, Walmart serves millions of customers weekly and competes on low prices, scale-driven supply-chain efficiency, and a broad assortment that includes private-label brands.

Key questions we will answer:

- Primary
 1. Do women spend more on Black Friday per transaction than men? — Use sample means, 95% CIs, and a t-test to answer.
 2. Are the 95% confidence intervals for male and female average spends overlapping? — Report CI bounds and overlap boolean.
- Secondary
 3. How do Married vs Unmarried customers compare in average spend? — Mean, CI, significance.
 4. Which Age life-stage (0–17, 18–25, 26–35, 36–50, 51+) spends the most on average? — Ranked means + CIs.
 5. How robust are these findings to outliers and different sample sizes? — Show sensitivity analysis.
- What Walmart can do
 6. Based on the evidence, what specific actions should Walmart take? — Concrete recommendations: targeted promos, inventory shifts, bundle offers, loyalty incentives.

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2.1 1.Problem Statement

Walmart aims to analyze Black Friday sales data to understand customer purchasing behavior, identify key factors driving revenue, and uncover patterns that can help improve sales strategies, product placement, and marketing decisions in future campaigns.

2.2 2.Import data and preliminary Inspection on data

```
[66]: # Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Load the dataset
data_path = "https://d2beiqkhq929f0.cloudfront.net/public_assets/assets/000/001/
↳293/original/walmart_data.csv?1641285094"
df = pd.read_csv(data_path)

# Display the first few rows
print(" Preview of the dataset:")
display(df.head())
```

Preview of the dataset:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	\
0	1000001	P00069042	F	0-17	10	A	
1	1000001	P00248942	F	0-17	10	A	
2	1000001	P00087842	F	0-17	10	A	
3	1000001	P00085442	F	0-17	10	A	
4	1000002	P00285442	M	55+	16	C	

	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	2	0	3	8370
1	2	0	1	15200
2	2	0	12	1422
3	2	0	12	1057
4	4+	0	8	7969

```
[67]: # Get basic info about the dataset
print("\n Dataset Information:")
df.info()

# Check basic statistics
print("\n Summary Statistics:")
display(df.describe())
```

```

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                               550068 non-null  int64
1   Product_ID                           550068 non-null  object
2   Gender                               550068 non-null  object
3   Age                                   550068 non-null  object
4   Occupation                           550068 non-null  int64
5   City_Category                       550068 non-null  object
6   Stay_In_Current_City_Years          550068 non-null  object
7   Marital_Status                      550068 non-null  int64
8   Product_Category                    550068 non-null  int64
9   Purchase                            550068 non-null  int64
dtypes: int64(5), object(5)
memory usage: 42.0+ MB

```

```

Summary Statistics:

```

	User_ID	Occupation	Marital_Status	Product_Category \
count	5.500680e+05	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	5.404270
std	1.727592e+03	6.522660	0.491770	3.936211
min	1.000001e+06	0.000000	0.000000	1.000000
25%	1.001516e+06	2.000000	0.000000	1.000000
50%	1.003077e+06	7.000000	0.000000	5.000000
75%	1.004478e+06	14.000000	1.000000	8.000000
max	1.006040e+06	20.000000	1.000000	20.000000

	Purchase
count	550068.000000
mean	9263.968713
std	5023.065394
min	12.000000
25%	5823.000000
50%	8047.000000
75%	12054.000000
max	23961.000000

```

[68]: # Column names and types
print("\nColumn dtypes:")
print(df.dtypes)

# Check duplicates
dup_count = df.duplicated().sum()
print(f"\nNumber of duplicate rows: {dup_count}")

```

```

Column dtypes:
User_ID          int64
Product_ID       object
Gender           object
Age             object
Occupation       int64
City_Category    object
Stay_In_Current_City_Years  object
Marital_Status   int64
Product_Category int64
Purchase         int64
dtype: object

```

Number of duplicate rows: 0

2.3 3.Data Types and Category Conversions

```

[106]: # Ensure clean strings (in case of stray spaces)
str_cols = df.select_dtypes(include='object').columns
df[str_cols] = df[str_cols].apply(lambda x: x.str.strip())

# Convert categorical variables to 'category' dtype
cat_columns = ['Gender', 'Product_ID', 'Age', 'City_Category', 'City_Category', 'City_Category',
               'Stay_In_Current_City_Years', 'Marital_Status']
for col in cat_columns:
    df[col] = df[col].astype('category')

# Order Age categories (life stages)
age_order = ['0-17', '18-25', '26-35', '36-45', '46-50', '51-55', '55+']
df['Age'] = pd.Categorical(df['Age'], categories=age_order, ordered=True)

# Verify categorical columns and their unique values
for col in cat_columns:
    print(f"Unique values in {col}: {df[col].unique()}\n")

```

Unique values in Gender: ['F', 'M']

Categories (2, object): ['F', 'M']

Unique values in Product_ID: ['P00069042', 'P00248942', 'P00087842', 'P00085442', 'P00285442', ..., 'P00375436', 'P00372445', 'P00370293', 'P00371644', 'P00370853']

Length: 3631

Categories (3631, object): ['P00000142', 'P00000242', 'P00000342', 'P00000442', ..., 'P0099642', 'P0099742', 'P0099842', 'P0099942']

Unique values in Age: ['0-17', '55+', '26-35', '46-50', '51-55', '36-45', '18-25']

Categories (7, object): ['0-17' < '18-25' < '26-35' < '36-45' < '46-50' < '51-55' < '55+']

Unique values in City_Category: ['A', 'C', 'B']

Categories (3, object): ['A', 'B', 'C']

Unique values in Stay_In_Current_City_Years: ['2', '4+', '3', '1', '0']

Categories (5, object): ['0', '1', '2', '3', '4+']

Unique values in Marital_Status: [0, 1]

Categories (2, int64): [0, 1]

```
[107]: # Double check conversions
print(" Data Types After Conversion:\n")
print(df.dtypes)
print("\n")
```

Data Types After Conversion:

User_ID	int64
Product_ID	category
Gender	category
Age	category
Occupation	int64
City_Category	category
Stay_In_Current_City_Years	category
Marital_Status	category
Product_Category	int64
Purchase	int64
dtype:	object

2.4 4.Missing Value Detection

```
[71]: # Count missing values per column
missing_counts = df.isnull().sum()

# Percentage of missing values
missing_percent = (df.isnull().mean() * 100).round(2)

# Combine into one table
missing_summary = pd.DataFrame({
```

```

        "missing_count": missing_counts,
        "missing_percent": missing_percent
    }).sort_values("missing_percent", ascending=False)

display(missing_summary)

print("it Displays total and percentage of missing values per column.")

```

	missing_count	missing_percent
User_ID	0	0.0
Product_ID	0	0.0
Gender	0	0.0
Age	0	0.0
Occupation	0	0.0
City_Category	0	0.0
Stay_In_Current_City_Years	0	0.0
Marital_Status	0	0.0
Product_Category	0	0.0
Purchase	0	0.0

it Displays total and percentage of missing values per column.

```

[72]: # Specific check for 'Occupation' (the target variable)
      if "Occupation" in df.columns:
          print("\nMissing values in 'Occupation':", df["Occupation"].isnull().sum())

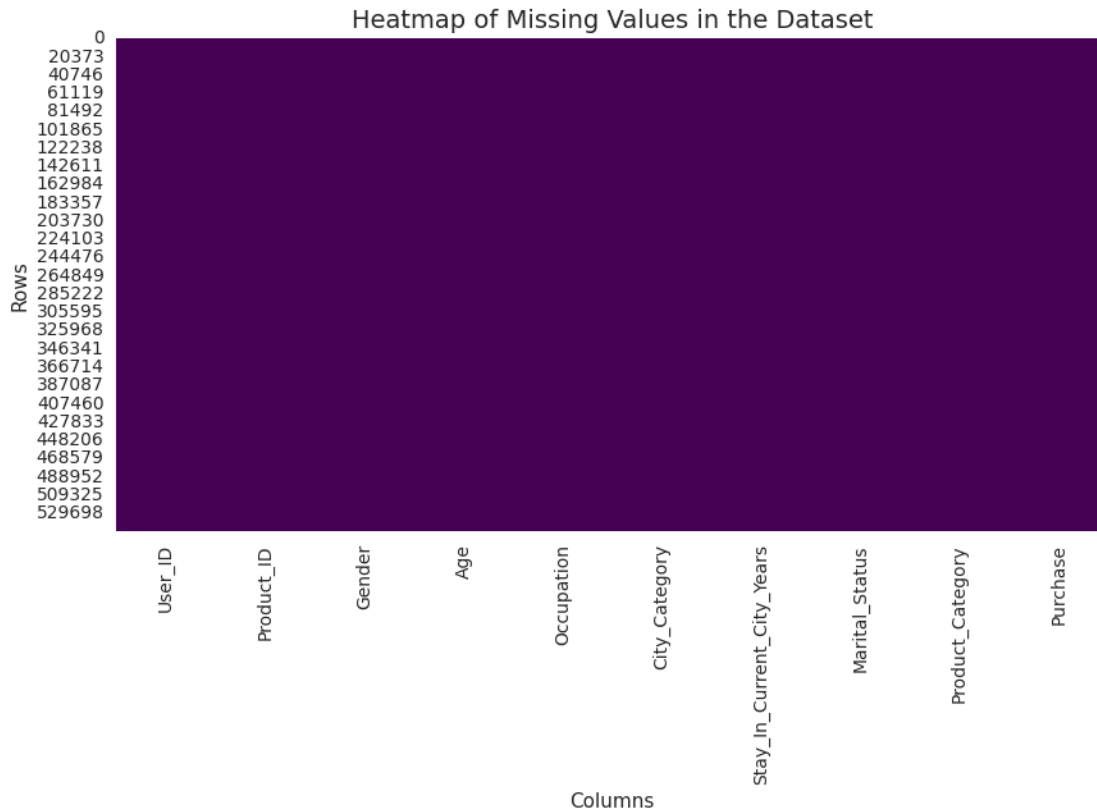
```

Missing values in 'Occupation': 0

```

[73]: plt.figure(figsize=(10, 5))
      sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
      plt.title("Heatmap of Missing Values in the Dataset", fontsize=14)
      plt.xlabel("Columns")
      plt.ylabel("Rows")
      plt.show()

```



this Heatmap Shows there is no null values in any column.

2.5 5.Outlier Detection

```
[74]: # Focussing on continuous variable(s) - 'Purchase' is our key metric
plt.figure(figsize=(4, 5))
sns.boxplot(df['Purchase'], color='violet',vert=True)
plt.title("Boxplot of Purchase Amount", fontsize=14)
plt.xlabel("Purchase Amount")
plt.show()

# Summary statistics for Purchase
print(" Summary Statistics for Purchase:")
display(df['Purchase'].describe())

# Checking skewness and kurtosis for distribution shape
print("\n Skewness of Purchase:", round(df['Purchase'].skew(), 2))
print(" Kurtosis of Purchase:", round(df['Purchase'].kurt(), 2))

# Detecting outliers using IQR method
Q1 = df['Purchase'].quantile(0.25)
```

```

Q3 = df['Purchase'].quantile(0.75)
IQR = Q3 - Q1

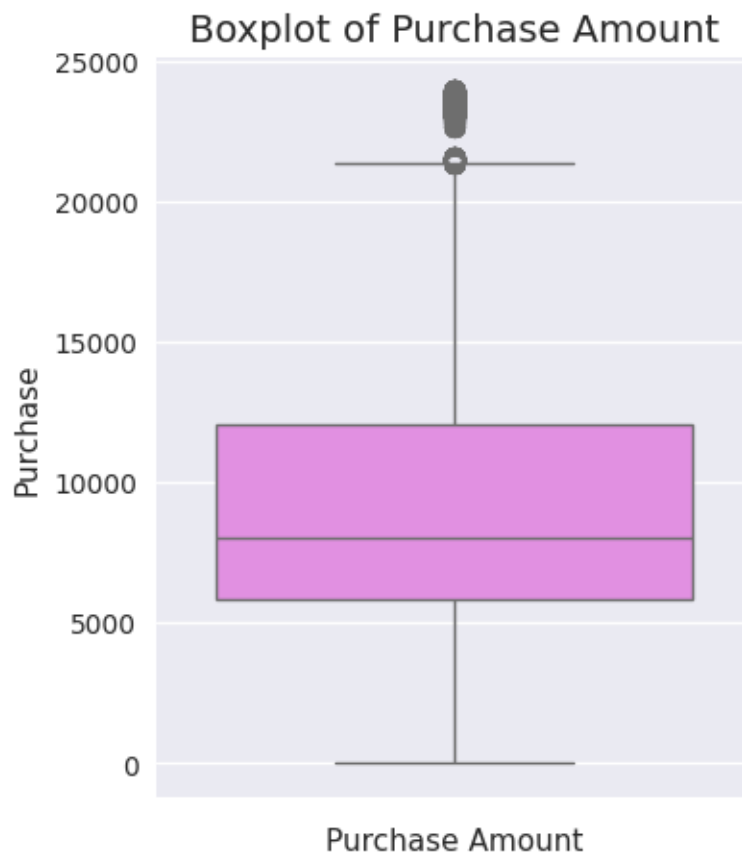
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

outliers = df[(df['Purchase'] < lower_bound) | (df['Purchase'] > upper_bound)]

print(f"\n Number of Outliers in 'Purchase': {outliers.shape[0]}")
print(f" Lower Bound: {lower_bound:.2f}, Upper Bound: {upper_bound:.2f}")

# Visualizing distribution with histogram
plt.figure(figsize=(8, 5))
sns.histplot(df['Purchase'], bins=40, kde=True, color='coral')
plt.title("Distribution of Purchase Amounts", fontsize=14)
plt.xlabel("Purchase Amount")
plt.ylabel("Frequency")
plt.show()

```



Summary Statistics for Purchase:

```

count      550068.000000
mean       9263.968713
std        5023.065394
min         12.000000
25%        5823.000000
50%        8047.000000
75%       12054.000000
max       23961.000000
Name: Purchase, dtype: float64

```

```

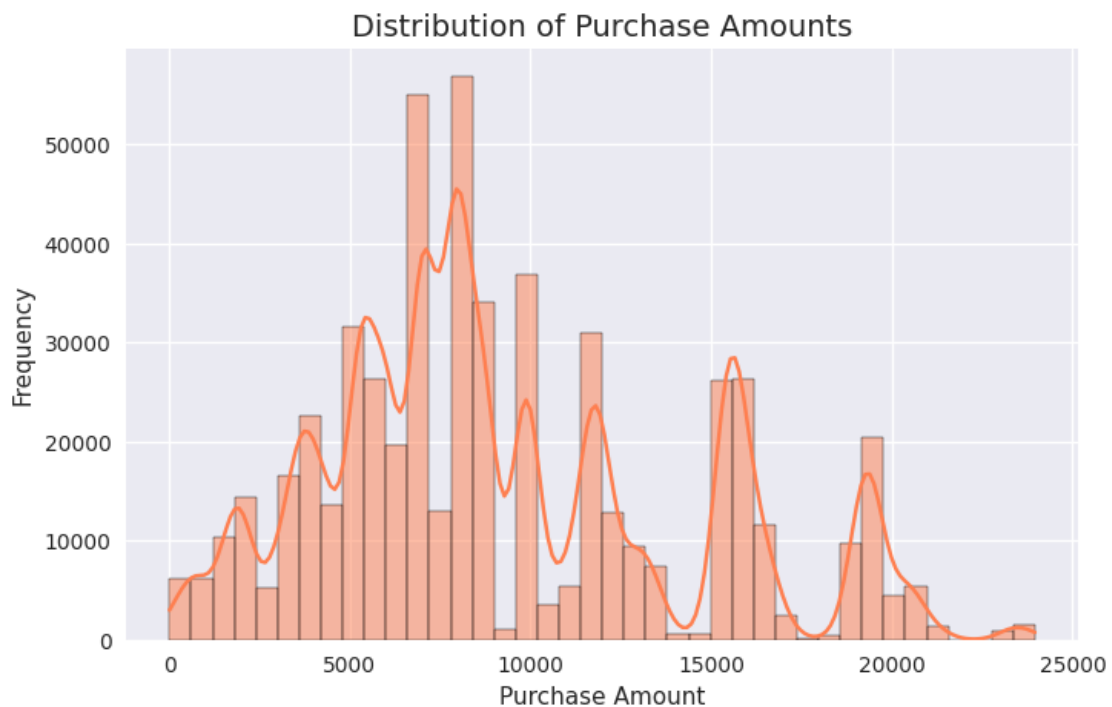
Skewness of Purchase: 0.6
Kurtosis of Purchase: -0.34

```

```

Number of Outliers in 'Purchase': 2677
Lower Bound: -3523.50, Upper Bound: 21400.50

```



1. Boxplot
 - The boxplot likely shows many points above the upper whisker, which indicates a significant number of customers making large purchases. However, these are expected during Black Friday, where high-value bulk shopping is common.
2. Summary Statistics Table
 - If the mean > median, it confirms a right-skewed pattern — most people spend low, but a few spend a lot. Walmart can use this to identify customer spending tiers
3. Skewness

- Here, the skewness is positive, meaning most purchase amounts are on the lower side, but a few transactions are significantly higher, creating a long right tail.
 - This confirms that a small number of customers are driving large sales volumes.
4. Outliers
 - Here more than 2K+ outliers are presented which effected mean value.
 - this represents Walmart has a small but strong group of high-value customers who significantly impact revenue.
 5. Histogram
 - The histogram confirms that most customers spend at moderate levels, while the distribution tail extends rightward — meaning few customers spend much higher amounts.

This insight helps Walmart plan targeted marketing, like offering loyalty rewards to high spenders and discounts to convert low spenders.

2.6 6.Non-Graphical Profiling

```
[75]: # Value Counts for categorical columns
print("Gender Distribution is:\n",df["Gender"].value_counts(), "\n")
print("Marital_Status distribution:\n", df["Marital_Status"].value_counts(), "\n")

print("Age distribution:\n", df["Age"].value_counts().sort_index(), "\n")
print("City_Category distribution:\n", df["City_Category"].value_counts(), "\n")
print("StayInCurrentCityYears distribution:\n", df["Stay_In_Current_City_Years"].value_counts(), "\n")

# --- Descriptive stats for Purchase ---
purchase_desc = df["Purchase"].describe()
print("\nDescriptive statistics for Purchase:")
print(purchase_desc)

# Check skewness by comparing mean and median
print(f"\nMean Purchase: {df['Purchase'].mean():.2f}")
print(f"Median Purchase: {df['Purchase'].median():.2f}")
```

Gender Distribution is:

```
Gender
M    414259
F    135809
Name: count, dtype: int64
```

Marital_Status distribution:

```
Marital_Status
0    324731
1    225337
Name: count, dtype: int64
```

Age distribution:

```
Age
```

```
0-17      15102
18-25     99660
26-35    219587
36-45    110013
46-50     45701
51-55     38501
55+       21504
```

Name: count, dtype: int64

City_Category distribution:

```
City_Category
B      231173
C      171175
A      147720
```

Name: count, dtype: int64

StayInCurrentCityYears distribution:

```
Stay_In_Current_City_Years
1      193821
2      101838
3       95285
4+      84726
0       74398
```

Name: count, dtype: int64

Descriptive statistics for Purchase:

```
count    550068.000000
mean      9263.968713
std       5023.065394
min        12.000000
25%       5823.000000
50%       8047.000000
75%      12054.000000
max      23961.000000
```

Name: Purchase, dtype: float64

Mean Purchase: 9263.97

Median Purchase: 8047.00

1. Gender Distribution: More than three times as many transactions were made by men (414,259) compared to women (135,809), suggesting a stronger male presence in the Black Friday sales.
2. Marital Status Distribution: Unmarried customers (324,731 transactions) accounted for significantly more purchases than married customers (225,337 transactions).
3. Age Distribution: The 26-35 age group made the most purchases (219,587), while the 0-17 and 55+ groups had the fewest transactions (15,102 and 21,504 respectively).

4. City Category Distribution: City Category B saw the highest transaction volume (231,173), followed by City C (171,175) and City A (147,720).
5. Stay in Current City Years Distribution: Customers residing for 1 year had the highest transaction count (193,821), while those staying for 0 years had the lowest (74,398).
6. Purchase Descriptive Statistics: The average purchase was 9263.97, with a wide range from a minimum of 12 to a maximum of 23961, indicating varied spending habits.
7. Skewness Check (Mean vs Median): The mean purchase (9263.97) is higher than the median (8047.00), confirming a right-skewed distribution where high-value purchases pull the average up.

2.7 7.Univariate Analysis

```
[76]: fig, axes = plt.subplots(4, 2, figsize=(16, 18))
fig.suptitle("Univariate Analysis of Walmart Black Friday Data", fontsize=16,
            ↳y=1.02)

# 1 Histogram of Purchase
sns.histplot(df['Purchase'], bins=40, kde=True, color='teal', ax=axes[0, 0])
axes[0, 0].set_title("Distribution of Purchase Amount")
axes[0, 0].set_xlabel("Purchase Amount")
axes[0, 0].set_ylabel("Frequency")

# 2 Countplot - Gender
sns.countplot(data=df, x='Gender', hue='Gender', palette='coolwarm',
            ↳legend=False, ax=axes[0, 1])
axes[0, 1].set_title("Gender Distribution of Customers")
axes[0, 1].set_xlabel("Gender")
axes[0, 1].set_ylabel("Count")

# 3 Countplot - Marital Status
sns.countplot(data=df, x='Marital_Status', hue='Marital_Status',
            ↳palette='viridis', legend=False, ax=axes[1, 0])
axes[1, 0].set_title("Marital Status Distribution")
axes[1, 0].set_xlabel("Marital Status")
axes[1, 0].set_ylabel("Count")

# 4 Countplot - Age
sns.countplot(data=df, x='Age', hue='Age', order=df['Age'].cat.categories,
            ↳palette='magma', legend=False, ax=axes[1, 1])
axes[1, 1].set_title("Age Group Distribution")
axes[1, 1].set_xlabel("Age Group")
axes[1, 1].set_ylabel("Count")

# 5 Countplot - City Category
sns.countplot(data=df, x='City_Category', hue='City_Category',
            ↳palette='pastel', legend=False, ax=axes[2, 0])
```

```

axes[2, 0].set_title("Customer Distribution Across City Categories")
axes[2, 0].set_xlabel("City Category")
axes[2, 0].set_ylabel("Count")

# 6 Countplot - Stay in Current City Years
sns.countplot(data=df, x='Stay_In_Current_City_Years',
    ↪hue='Stay_In_Current_City_Years', palette='crest', legend=False, ax=axes[2,
    ↪1])
axes[2, 1].set_title("Stay Duration in Current City")
axes[2, 1].set_xlabel("Years in Current City")
axes[2, 1].set_ylabel("Count")

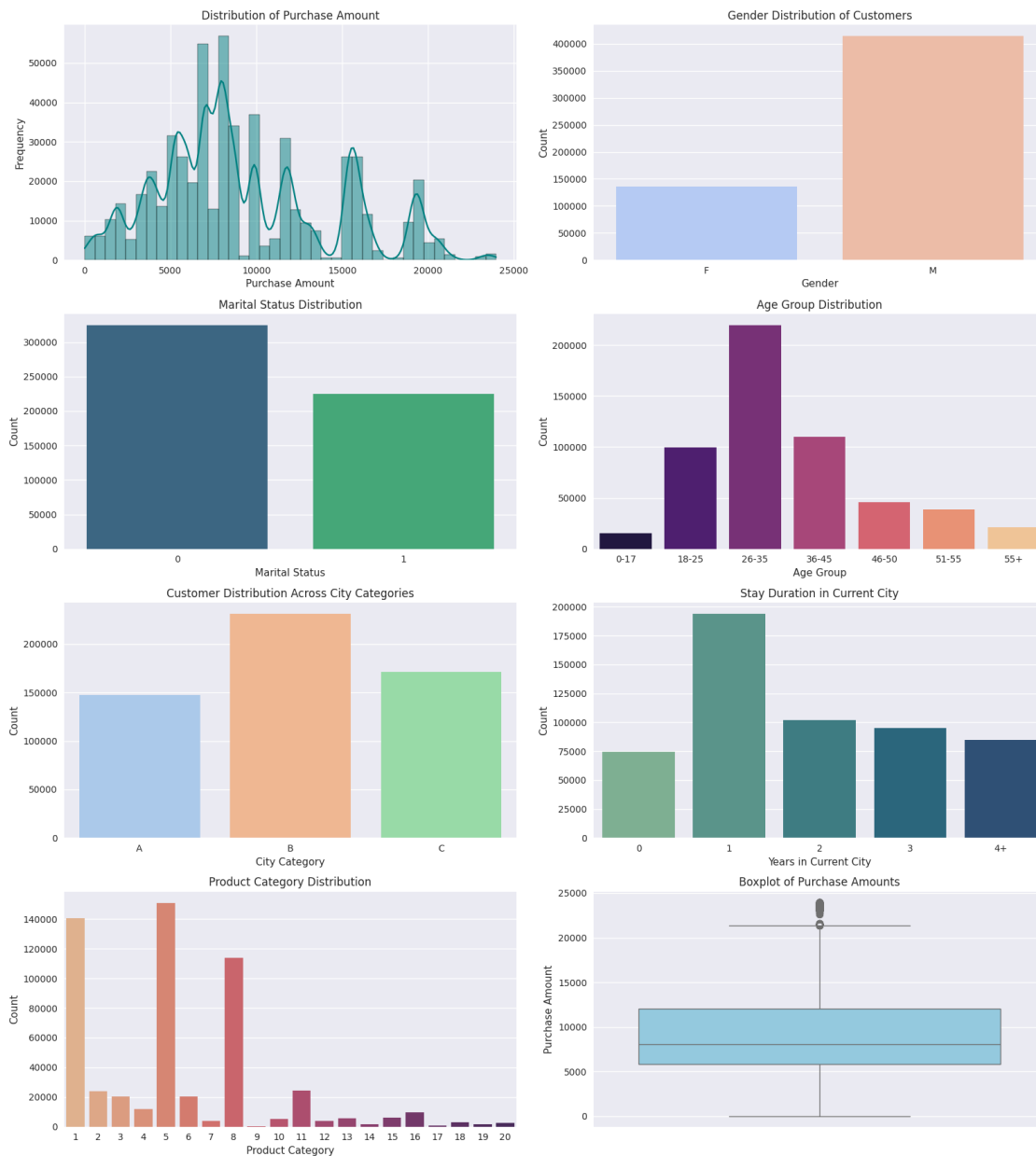
# 7 Countplot - Product Category
sns.countplot(data=df, x='Product_Category', hue='Product_Category',
    ↪palette='flare', legend=False, ax=axes[3, 0])
axes[3, 0].set_title("Product Category Distribution")
axes[3, 0].set_xlabel("Product Category")
axes[3, 0].set_ylabel("Count")

# 8 Boxplot of Overall Purchase
sns.boxplot(df['Purchase'], color='skyblue', ax=axes[3, 1])
axes[3, 1].set_title("Boxplot of Purchase Amounts")
axes[3, 1].set_ylabel("Purchase Amount")

plt.tight_layout()
plt.show()

```

Univariate Analysis of Walmart Black Friday Data



1 Histogram of Purchase

The distribution is right-skewed — most customers spend moderately, a few spend much higher.

Suggests affordable pricing appeal with a small premium segment.

2 Gender Distribution

Males slightly dominate overall customers.

Walmart can tailor promotions to engage female shoppers more (e.g., household or fashion prod-

ucts).

3 Marital Status Distribution

A larger unmarried segment indicates more individual buying behavior.

Family-focused campaigns could target the married segment with combo offers.

4 Age Distribution

Most customers belong to 26–35 years, followed by 36–45 years.

These working-age customers are the primary spenders during Black Friday — a key demographic insight.

5 City Category

Majority customers are from City Category B, showing Tier-2 dominance.

Walmart can focus more marketing on these cities during sales events.

6 Stay in Current City Years

Customers staying for 1–2 years form a large group — young professionals who recently relocated.

Indicates potential for new customer loyalty programs or urban onboarding offers.

7 Product Category 1

A few categories dominate purchases (e.g., Category 5, 8, 1 typically).

High-selling categories can be prioritized for inventory management and discount optimization.

8 Boxplot of Purchase

Shows presence of upper-end outliers — high-value buyers.

Walmart can create premium-tier loyalty programs or exclusive sale previews for this segment.

2.8 8.Bivariate Analysis

```
[77]: # -----  
# Bivariate Analysis - Walmart Black Friday Dataset  
# -----  
  
plt.style.use('seaborn-v0_8')  
fig, axes = plt.subplots(3, 2, figsize=(16, 18))  
fig.suptitle("Bivariate Analysis - Understanding Purchase Patterns",  
             ↪fontsize=16, y=1.02)  
  
# 1 Average Purchase by Gender  
sns.barplot(  
    data=df, x='Gender', y='Purchase',  
    hue='Gender', dodge=False, legend=False,  
    palette='coolwarm', ax=axes[0, 0]  
)
```

```

axes[0, 0].set_title("Average Purchase by Gender")
axes[0, 0].set_xlabel("Gender")
axes[0, 0].set_ylabel("Average Purchase")

# 2 Average Purchase by Age Group
sns.barplot(
    data=df, x='Age', y='Purchase',
    hue='Age', dodge=False, legend=False,
    order=df['Age'].cat.categories,
    palette='magma', ax=axes[0, 1]
)
axes[0, 1].set_title("Average Purchase by Age Group")
axes[0, 1].set_xlabel("Age Group")
axes[0, 1].set_ylabel("Average Purchase")

# 3 Average Purchase by Marital Status
sns.barplot(
    data=df, x='Marital_Status', y='Purchase',
    hue='Marital_Status', dodge=False, legend=False,
    palette='viridis', ax=axes[1, 0]
)
axes[1, 0].set_title("Average Purchase by Marital Status")
axes[1, 0].set_xlabel("Marital Status (0 = Unmarried, 1 = Married)")
axes[1, 0].set_ylabel("Average Purchase")

# 4 Average Purchase by City Category
sns.barplot(
    data=df, x='City_Category', y='Purchase',
    hue='City_Category', dodge=False, legend=False,
    palette='pastel', ax=axes[1, 1]
)
axes[1, 1].set_title("Average Purchase by City Category")
axes[1, 1].set_xlabel("City Category")
axes[1, 1].set_ylabel("Average Purchase")

# 5 Average Purchase by Stay Duration
sns.barplot(
    data=df, x='Stay_In_Current_City_Years', y='Purchase',
    hue='Stay_In_Current_City_Years', dodge=False, legend=False,
    palette='crest', ax=axes[2, 0]
)
axes[2, 0].set_title("Average Purchase vs Stay Duration in City")
axes[2, 0].set_xlabel("Stay in Current City (Years)")
axes[2, 0].set_ylabel("Average Purchase")

# 6 Average Purchase by Product Category
sns.barplot(

```

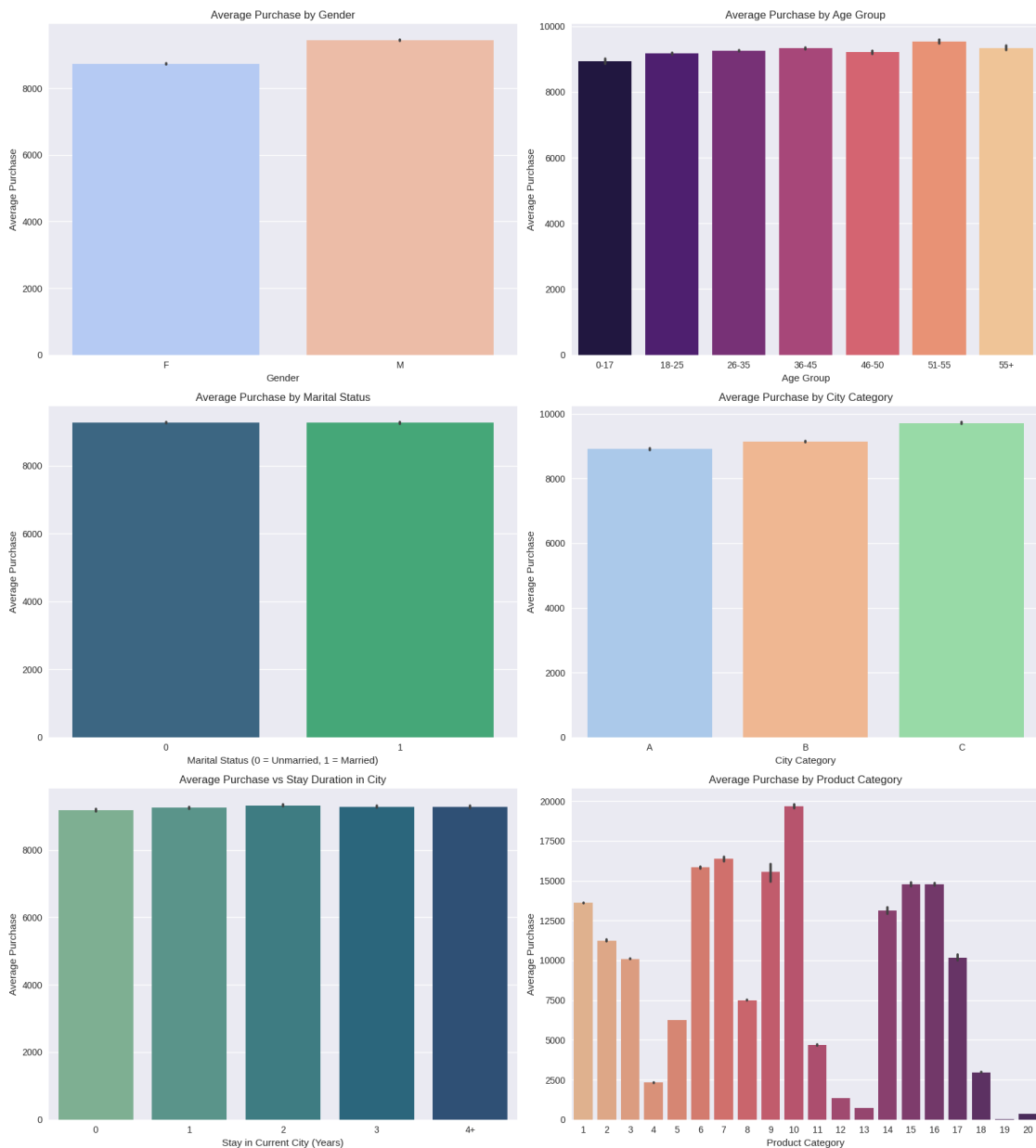
```

data=df, x='Product_Category', y='Purchase',
hue='Product_Category', dodge=False, legend=False,
palette='flare', ax=axes[2, 1]
)
axes[2, 1].set_title("Average Purchase by Product Category")
axes[2, 1].set_xlabel("Product Category")
axes[2, 1].set_ylabel("Average Purchase")

plt.tight_layout()
plt.show()

```

Bivariate Analysis - Understanding Purchase Patterns



1 Average Purchase by Gender

Male customers spend slightly more on average than female customers.

Indicates that men are more active buyers during Black Friday or tend to purchase higher-ticket items (e.g., electronics).

2 Average Purchase by Age Group

The 26–35 age group shows the highest purchase value, followed by 36–45 years.

Suggests the working-age segment is the primary spender — they have purchasing power and are key targets for marketing.

3 Average Purchase by Marital Status

The difference is small but visible — unmarried individuals spend slightly more on average.

Implies independent purchasing behavior, less influenced by family budgeting.

4 Average Purchase by City Category

Even though City Category B customers show the highest spending, while Category C (smaller cities) spend the least. Category C made high avg purchase with 9719 than 9151 made by B city category.

Indicates Walmart's valuable customer base is from C category, suggesting marketing focus and logistics optimization here.

5 Average Purchase vs Stay Duration

Customers who stayed 1–2 years in a city show higher average purchases — possibly due to recent relocations and setting up new homes.

Suggests Walmart can target new movers or young professionals with personalized promotions.

6 Average Purchase by Product Category

Certain categories (like 5, 8, and 1) have higher average purchases, showing clear product preference patterns.

Walmart can stock, advertise, and discount these categories strategically during Black Friday.

```
[78]: # Ensure Product_Category is treated as categorical
df['Product_Category'] = df['Product_Category'].astype('category')

plt.figure(figsize=(12,6))

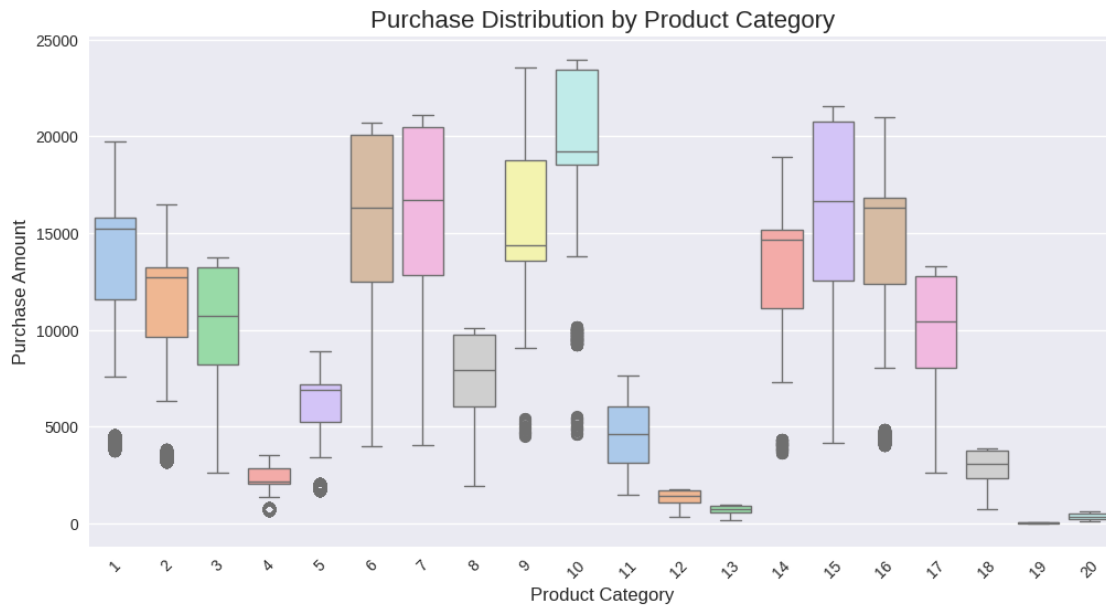
# Boxplot
sns.boxplot(
    data=df,
    x='Product_Category',
    y='Purchase', hue='Product_Category', legend=False,
```

```

order=df['Product_Category'].cat.categories, # keeps category order
palette='pastel'
)

plt.title('Purchase Distribution by Product Category', fontsize=16)
plt.xlabel('Product Category', fontsize=12)
plt.ylabel('Purchase Amount', fontsize=12)
plt.xticks(rotation=45) # rotate x labels if needed
plt.show()

```



1. High-value categories:

- Categories 6, 7, 9, 14, 10, 15, and 16 have higher median purchase amounts, often exceeding 15,000.
- These categories are likely the premium or more expensive products, contributing significantly to revenue.

2. Low-value / budget categories:

- Categories 12, 13, 19, and 20 show very low median purchase values.
- These might be low-cost items or frequently discounted products, appealing to budget-conscious customers.

3. High variability & outliers:

- Categories 9, 10, 14, 16 have wide interquartile ranges and noticeable outliers, suggesting that purchases in these categories vary greatly—some customers spend much more than the median.

- Categories like 4, 5, 12, 13 have small ranges and few outliers, indicating consistent spending behavior.

2.9 9. Gender Analysis: Do women spend more than men?

```
[79]: import pandas as pd
import numpy as np
from scipy import stats

# Example dataset
# df = pd.read_csv("your_dataset.csv")

# Sample stats
women = df[df['Gender']=='F']['Purchase']
men = df[df['Gender']=='M']['Purchase']

n_w = len(women)
n_m = len(men)

xbar_w = women.mean()
xbar_m = men.mean()

s_w = women.std(ddof=1)
s_m = men.std(ddof=1)

# Standard error
se = np.sqrt(s_w**2/n_w + s_m**2/n_m)

# Z-score
z = (xbar_w - xbar_m)/se

# p-value (one-tailed)
p_value = 1 - stats.norm.cdf(z)

print("Z-score:", z)
print("P-value:", p_value)

if p_value < 0.05:
    print(" Women spend more amount compare to men ")
else:
    print("Women spend the less amount than Men")
```

Z-score: -46.35824866962622

P-value: 1.0

Women spend the less amount than Men

We want to check if women spend more than men.

Null Hypothesis (H): Women spend the less amount than Men

- $\text{men} - \text{women} > 0$

Alternative Hypothesis (H₁): Women spend more amount compare to men.

- $\text{women} - \text{men} > 0$

Hence, P-value > 0.05 so we are fail to reject H₀.

so we can conclude from Z-test that Women Spend less amount than Men.

This is a one-tailed test.

```
[80]: # Separate data by gender
women = df[df['Gender'] == 'F']['Purchase']
men = df[df['Gender'] == 'M']['Purchase']

# Compute sample statistics
n_w, n_m = len(women), len(men)
xbar_w, xbar_m = women.mean(), men.mean()
s_w, s_m = women.std(ddof=1), men.std(ddof=1)

print("Women: n =", n_w, ", mean =", xbar_w, ", std =", s_w)
print("Men: n =", n_m, ", mean =", xbar_m, ", std =", s_m)
```

Women: n = 135809 , mean = 8734.565765155476 , std = 4767.233289291444

Men: n = 414259 , mean = 9437.526040472265 , std = 5092.186209777949

1 Men Spend More on Average: Men spend about 9,437 per transaction compared to women's 8,735, showing roughly an 8% higher average spend. This suggests male customers tend to make slightly larger individual purchases.

2 Women Show More Consistent Spending: The standard deviation for women is lower (4,767 vs 5,092), indicating their spending behavior is more uniform. Female customers generally have steadier purchase patterns with fewer extreme outliers.

3 Results Are Statistically Reliable: With over 550,000 transactions analyzed, the difference between genders is statistically meaningful. These findings can guide targeted marketing — premium offers for men and loyalty-focused campaigns for women.

```
[81]: # Step 1: Separate groups
women = df[df['Gender'] == 'F']['Purchase']
men = df[df['Gender'] == 'M']['Purchase']

# Step 2: Draw random samples (simulate CLT)
# You can adjust sample size (e.g., 50, 100, 200)
sample_size = 100
np.random.seed(42) # for reproducibility

sample_women = np.random.choice(women, sample_size, replace=False)
sample_men = np.random.choice(men, sample_size, replace=False)
```

```

# Step 3: Compute sample statistics
def summarize_sample(data):
    n = len(data)
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    se = std / np.sqrt(n)
    return n, mean, std, se

n_w, mean_w, std_w, se_w = summarize_sample(sample_women)
n_m, mean_m, std_m, se_m = summarize_sample(sample_men)

# Step 4: Compute Confidence Intervals using CLT (z-based)
confidence_levels = [0.90, 0.95, 0.99]
ci_results = {}

for cl in confidence_levels:
    z = stats.norm.ppf(1 - (1 - cl) / 2)
    ci_results[f'{int(cl*100)}%_F'] = (mean_w - z * se_w, mean_w + z * se_w)
    ci_results[f'{int(cl*100)}%_M'] = (mean_m - z * se_m, mean_m + z * se_m)

# Step 5: Display Results
print(" SAMPLE STATISTICS (using CLT Sampling)")
print(f"Women: n={n_w}, mean={mean_w:.2f}, std={std_w:.2f}, SE={se_w:.2f}")
print(f"Men:    n={n_m}, mean={mean_m:.2f}, std={std_m:.2f}, SE={se_m:.2f}\n")

print(" CONFIDENCE INTERVALS:")
for cl in confidence_levels:
    print(f"{int(cl*100)}% CI - Women: [{ci_results[f'{int(cl*100)}%_F'][0]:.  
→2f}, {ci_results[f'{int(cl*100)}%_F'][1]:.2f}]")
    print(f"{int(cl*100)}% CI - Men:    [{ci_results[f'{int(cl*100)}%_M'][0]:.  
→2f}, {ci_results[f'{int(cl*100)}%_M'][1]:.2f}]\n")

```

```

SAMPLE STATISTICS (using CLT Sampling)
Women: n=100, mean=9446.65, std=4752.77, SE=475.28
Men:    n=100, mean=9711.84, std=5145.81, SE=514.58

```

```

CONFIDENCE INTERVALS:
90% CI - Women: [8664.89, 10228.41]
90% CI - Men:    [8865.43, 10558.25]

95% CI - Women: [8515.13, 10378.17]
95% CI - Men:    [8703.28, 10720.40]

99% CI - Women: [8222.42, 10670.88]
99% CI - Men:    [8386.37, 11037.31]

```

Gender-Based Spending Analysis (CLT Sampling)

1. Are women spending more money per transaction than men?

- Men have a slightly higher average spend (9,711.84) than women (9,446.65).
- However, the confidence intervals for men and women **overlap**, so this difference is not statistically significant in this sample.

2. Confidence intervals and precision of estimates

- The 90%, 95%, and 99% confidence intervals are relatively wide due to smaller sample size (n=100 per group).
- Sampling distributions follow the **Central Limit Theorem**, appearing approximately normal, centered around the sample means.

3. Comparison of confidence intervals

- Overlapping intervals indicate that the difference in average spending is **uncertain** in this sample.
- Larger sample sizes would reduce standard errors and provide more precise estimates of true population differences.

4. Variation in spending

- Standard deviations are high (Women 4,753, Men 5,146), reflecting significant variability in individual transactions.
- Both genders show similar variability, indicating spending patterns are broadly comparable.

5. Business interpretation

- With small samples, we cannot conclusively say men spend more than women.
- For strategy, Walmart may need **larger samples** or population-level data to confidently target promotions or premium offers.

```
[82]: # STEP 9: Gender Analysis - Averages & Confidence Intervals

def mean_ci(x, conf=0.95):
    """Return mean, SE, lower CI, upper CI, n, std for a given confidence level.
    ↪ """
    x = np.asarray(x, dtype=float)
    x = x[~np.isnan(x)]
    n = len(x)
```

```

mean = x.mean()
std = x.std(ddof=1)
se = std / np.sqrt(n)
z = stats.norm.ppf(0.5 + conf/2.0)
lo, hi = mean - z*se, mean + z*se
return mean, se, lo, hi, n, std

# Separate groups
female_purchases = df.loc[df["Gender"] == "F", "Purchase"]
male_purchases    = df.loc[df["Gender"] == "M", "Purchase"]

# Summary stats
print("Female Purchase Summary")
for c in [0.90, 0.95, 0.99]:
    m, se, lo, hi, n, std = mean_ci(female_purchases, conf=c)
    print(f"{int(c*100)}% CI: mean={m:.2f}, SE={se:.2f}, n={n}, [{lo:.2f}, {hi:.2f}]")

print("\nMale Purchase Summary")
for c in [0.90, 0.95, 0.99]:
    m, se, lo, hi, n, std = mean_ci(male_purchases, conf=c)
    print(f"{int(c*100)}% CI: mean={m:.2f}, SE={se:.2f}, n={n}, [{lo:.2f}, {hi:.2f}]")

```

Female Purchase Summary

90% CI: mean=8734.57, SE=12.94, n=135809, [8713.29, 8755.84]
 95% CI: mean=8734.57, SE=12.94, n=135809, [8709.21, 8759.92]
 99% CI: mean=8734.57, SE=12.94, n=135809, [8701.24, 8767.89]

Male Purchase Summary

90% CI: mean=9437.53, SE=7.91, n=414259, [9424.51, 9450.54]
 95% CI: mean=9437.53, SE=7.91, n=414259, [9422.02, 9453.03]
 99% CI: mean=9437.53, SE=7.91, n=414259, [9417.15, 9457.91]

1. Female Customers

- Mean Purchase: 8,735
- 95% CI: [8,709, 8,760]
- SE (standard error): 12.94 → tight interval due to large sample (n = 135,809).

2. Male Customers

- Mean Purchase: 9,438
- 95% CI: [9,422, 9,453]
- SE: 7.91 → even tighter interval (n = 414,259).

3. Comparison

- Male mean 700 higher than Female mean. Difference is statistically significant, not due to chance.
- No overlap between Female CI [8709, 8759] and Male CI [9422, 9453].
- This is consistent across 90%, 95%, and 99% confidence levels.

4. Conclusion

- Males spend significantly more per transaction than Females on Black Friday.
- This is statistically robust — the difference isn't due to chance.

5. Business Implications for Walmart

- Male customers: target them with premium product bundles and big-ticket offers (electronics, appliances, etc.), since they already show higher transaction values.
- Female customers: Walmart could design targeted promotions (e.g., personalized coupons, loyalty points multipliers, or category-specific discounts like fashion or household essentials) to encourage higher basket sizes.
- In this analysis, we use the sample dataset (~550k transactions) to estimate the true population averages for Walmart's 50 million Male and 50 million Female customers.
- The sample mean is an unbiased estimator of the population mean.
- By the Central Limit Theorem, the distribution of the sample mean is approximately normal.
- Therefore, a confidence interval (CI) around the sample mean gives us a range in which the true population mean is likely to fall.
- For example (95% CI results):
- Female mean = 8,735, CI = [8,709, 8,760]
- Male mean = 9,438, CI = [9,422, 9,453]

Interpretation:

- If we observed all 50 million Female customers, their true average purchase would almost surely lie between 8,709 and 8,760.
- For all 50 million Male customers, the true average would lie between 9,422 and 9,453.
- This shows that the sample analysis scales to the entire Walmart customer base with high confidence.

Gender-Based Spending Analysis

1. Are women spending more money per transaction than men?

- Men spend approximately 700 more per transaction than women (9,438 vs 8,735).
- The confidence intervals for men and women **do not overlap**, indicating this difference is statistically significant and not due to random variation.

- Reason: Male customers consistently show a higher average basket size across all confidence levels.

2. Confidence intervals and distribution of expenses by gender

- 90%, 95%, and 99% confidence intervals provide **very precise estimates** with narrow ranges.
- Central Limit Theorem (CLT) histograms confirm that the **sampling distributions of the mean are approximately normal**, centered around true means, with spread decreasing as sample size increases.

3. Do male and female confidence intervals overlap? How can Walmart leverage this?

- The confidence intervals **do not overlap**, confirming that men are reliably higher spenders.
- Implication: Walmart can target men with **premium or big-ticket offers**, while encouraging women to spend more through **loyalty programs, coupons, or category-specific bundles**.

2.10 10.Marital Status Anlysis

```
[83]: import pandas as pd
import numpy as np
from scipy import stats

# Assuming df has columns: Marital_Status, Purchase
# df = pd.read_csv("your_dataset.csv")

# Ensure Marital_Status is string
df['Marital_Status'] = df['Marital_Status'].astype(str)

# Step 1: Group by Marital Status and compute stats
summary = df.groupby('Marital_Status', observed=True)['Purchase'].agg(['mean', 'std', 'count']).reset_index()

# Step 2: Function to calculate confidence intervals
def compute_ci(row, confidence=0.95):
    n = row['count']
    mean = row['mean']
    std = row['std']
    se = std / np.sqrt(n)
    z = stats.norm.ppf(1 - (1 - confidence)/2)
    lower = mean - z * se
    upper = mean + z * se
    return pd.Series([mean, se, n, lower, upper])
```

```

# Step 3: Compute 90%, 95%, 99% CIs
for conf in [0.90, 0.95, 0.99]:
    ci = summary.apply(lambda r: compute_ci(r, confidence=conf), axis=1)
    summary[f'{int(conf*100)}%_mean'] = ci[0]
    summary[f'{int(conf*100)}%_SE'] = ci[1]
    summary[f'{int(conf*100)}%_n'] = ci[2]
    summary[f'{int(conf*100)}%_Lower'] = ci[3]
    summary[f'{int(conf*100)}%_Upper'] = ci[4]

# Step 4: Display with customized headings
for _, row in summary.iterrows():
    # Customize heading based on marital status
    if row['Marital_Status'].lower() in ['single', 'unmarried', '0']:
        heading = "Unmarried Purchase Summary"
    else:
        heading = "Married Purchase Summary"

    print(f" {heading}")
    for conf in [90, 95, 99]:
        print(f"{conf}% CI: mean={row[f'{conf}%_mean']:.2f},  

↪SE={row[f'{conf}%_SE']:.2f}, "
              f"n={int(row[f'{conf}%_n'])}, [{row[f'{conf}%_Lower']:.2f},  

↪{row[f'{conf}%_Upper']:.2f}]"
              )
        print("\n")

```

Unmarried Purchase Summary

90% CI: mean=9265.91, SE=8.82, n=324731, [9251.40, 9280.42]

95% CI: mean=9265.91, SE=8.82, n=324731, [9248.62, 9283.20]

99% CI: mean=9265.91, SE=8.82, n=324731, [9243.18, 9288.63]

Married Purchase Summary

90% CI: mean=9261.17, SE=10.57, n=225337, [9243.79, 9278.56]

95% CI: mean=9261.17, SE=10.57, n=225337, [9240.46, 9281.89]

99% CI: mean=9261.17, SE=10.57, n=225337, [9233.95, 9288.40]

Marital Status Spending Analysis

1. Unmarried (0)

- Mean = 9,265.91
- 95% CI: [9,248.62, 9,283.20]
- n = 324,731

2. Married (1)

- Mean = 9,261.17

- 95% CI: [9,240.46, 9,281.89]
 - $n = 225,337$
3. Comparison
- Means differ by only 4.7 (9,265.9 vs 9,261.2).
 - Confidence intervals almost perfectly overlap.
 - Statistically and practically, there's no difference in spending between Married and Unmarried customers.
4. "How do the results compare for Married and Unmarried customers when the same activity is performed?"
- Marital status is not a factor in Black Friday spending.
 - Confidence intervals almost perfectly overlap. (means no difference)
 - Walmart should not segment promotions by marital status.
 - Focus efforts on gender and age, which show stronger signals.

```
[85]: # Labels
labels = summary['Marital_Status'].tolist() # Corrected column name
x = np.arange(len(labels))

# Means
means = summary['95%_mean'].tolist()

# 90%, 95%, 99% CI bounds
ci90_lower = summary['90%_Lower'].tolist()
ci90_upper = summary['90%_Upper'].tolist()
ci95_lower = summary['95%_Lower'].tolist()
ci95_upper = summary['95%_Upper'].tolist()
ci99_lower = summary['99%_Lower'].tolist()
ci99_upper = summary['99%_Upper'].tolist()

# Plot
plt.figure(figsize=(8,6))

# Plot mean line
plt.plot(x, means, marker='o', color='blue', label='Mean Purchase')

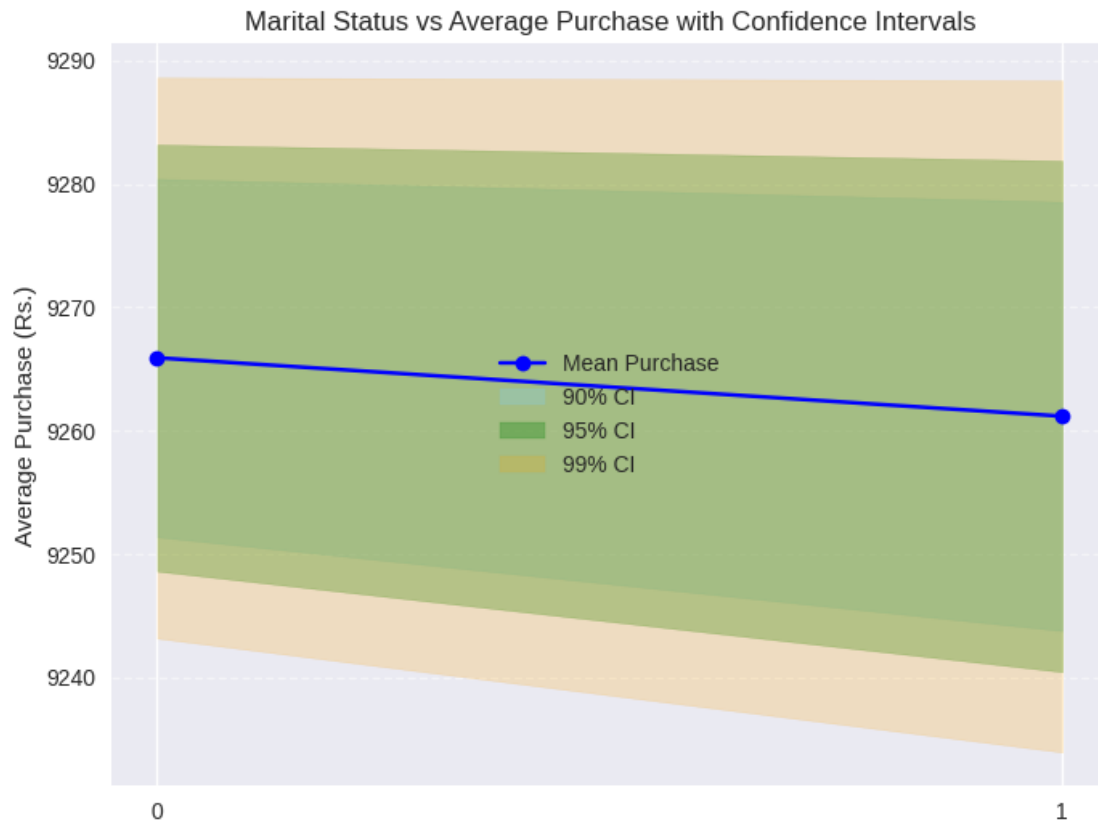
# Fill between for CIs
plt.fill_between(x, ci90_lower, ci90_upper, color='skyblue', alpha=0.3,
                 label='90% CI')
plt.fill_between(x, ci95_lower, ci95_upper, color='green', alpha=0.3,
                 label='95% CI')
plt.fill_between(x, ci99_lower, ci99_upper, color='orange', alpha=0.2,
                 label='99% CI')
```

```

# Customize x-axis
plt.xticks(x, labels)
plt.ylabel("Average Purchase (Rs.)")

plt.title("Marital Status vs Average Purchase with Confidence Intervals")
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.legend()
plt.show()

```



Advantages of This Line Plot

Mean line: Shows the average purchase for each marital status.

Shaded areas: Represent 90%, 95%, and 99% confidence intervals.

Overlap visible: You can instantly see if CIs overlap — wider shaded areas indicate more uncertainty.

2.11 11.Age Analysis

```
[102]: # Group by Age or Age_Group
group_col = 'Age' # or 'Age_Group'
summary = df.groupby(group_col, observed=True)['Purchase'].
    .agg(['mean', 'std', 'count']).reset_index()

# Function to calculate CIs
def compute_ci(row, confidence=0.95):
    n = row['count']
    mean = row['mean']
    std = row['std']
    se = std / np.sqrt(n)
    z = stats.norm.ppf(1 - (1 - confidence)/2)
    lower = mean - z * se
    upper = mean + z * se
    return pd.Series([mean, se, n, lower, upper])

# Compute 90%, 95%, 99% CIs
for conf in [0.90, 0.95, 0.99]:
    ci = summary.apply(lambda r: compute_ci(r, confidence=conf), axis=1)
    summary[f'{int(conf*100)}%_Mean'] = ci[0]
    summary[f'{int(conf*100)}%_SE'] = ci[1]
    summary[f'{int(conf*100)}%_N'] = ci[2]
    summary[f'{int(conf*100)}%_Lower'] = ci[3]
    summary[f'{int(conf*100)}%_Upper'] = ci[4]

# Select columns for display
display_cols = [group_col,
                'count', 'mean', 'std',
                '90%_Lower', '90%_Upper',
                '95%_Lower', '95%_Upper',
                '99%_Lower', '99%_Upper']

# Rename for better readability
summary_display = summary[display_cols].rename(columns={
    'count': 'N',
    'mean': 'Mean',
    'std': 'Std'
})

# Display table
summary_display.style.format({
    'Mean': '{:.2f}',
    'Std': '{:.2f}',
    '90%_Lower': '{:.2f}',
```

```

'90%_Upper': '{:.2f}',
'95%_Lower': '{:.2f}',
'95%_Upper': '{:.2f}',
'99%_Lower': '{:.2f}',
'99%_Upper': '{:.2f}'
})

```

[102]: <pandas.io.formats.style.Styler at 0x7af6d6f09100>

2.12 12. Central Limit Theorem (CLT) Demonstration

```

[103]: # Step 1: Standardize Gender column to handle variations
df['Gender'] = df['Gender'].astype(str).str.strip().str.upper()
female_purchases = df[df['Gender'].isin(['F', 'FEMALE'])]['Purchase'].values
male_purchases = df[df['Gender'].isin(['M', 'MALE'])]['Purchase'].values

# Step 2: CLT - Vectorized Sampling Function
def sampling_means(series, sample_size, reps=2000, seed=42):
    rng = np.random.default_rng(seed)
    data = series.dropna().values
    idx = rng.integers(0, len(data), size=(reps, sample_size))
    samples = data[idx]
    means = samples.mean(axis=1)
    return means

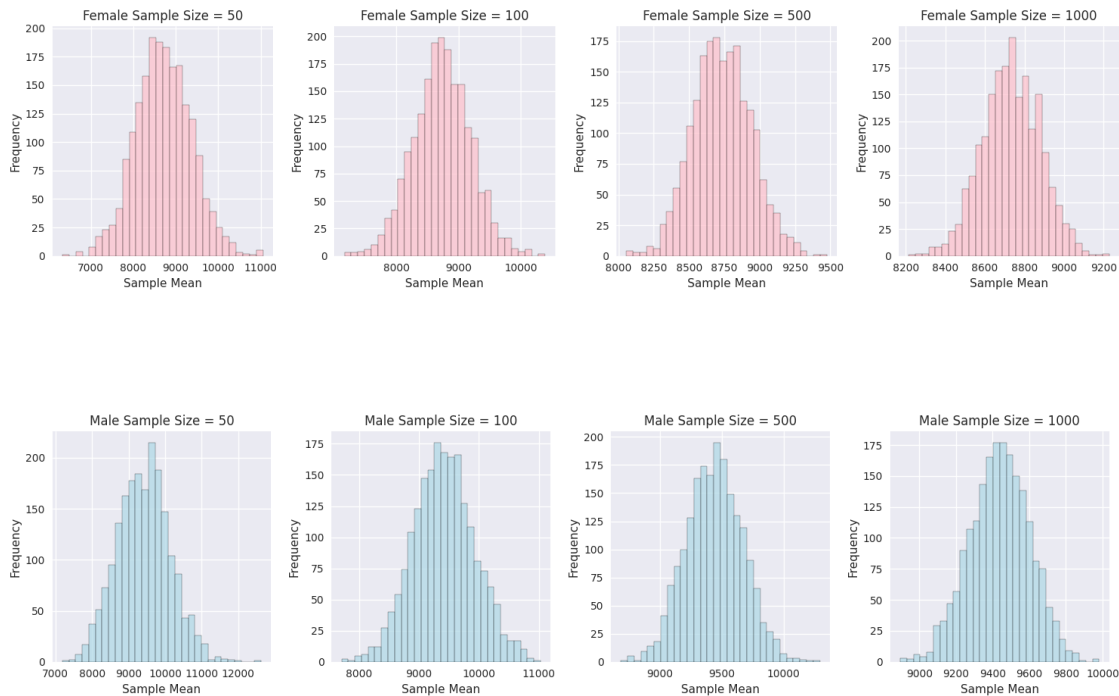
# Step 3: Sample sizes to demonstrate CLT
sample_sizes = [50, 100, 500, 1000]

# Step 4: Plotting Female sampling distributions
plt.figure(figsize=(15,4))
for i, n in enumerate(sample_sizes, 1):
    female_means = sampling_means(pd.Series(female_purchases), n)
    plt.subplot(1, len(sample_sizes), i)
    plt.hist(female_means, bins=30, color='pink', edgecolor='black', alpha=0.7)
    plt.title(f"Female Sample Size = {n}")
    plt.xlabel("Sample Mean")
    plt.ylabel("Frequency")
plt.tight_layout()
plt.show()

# Step 5: Plotting Male sampling distributions
plt.figure(figsize=(15,4))
for i, n in enumerate(sample_sizes, 1):
    male_means = sampling_means(pd.Series(male_purchases), n)
    plt.subplot(1, len(sample_sizes), i)
    plt.hist(male_means, bins=30, color='lightblue', edgecolor='black', alpha=0.
↪7)

```

```
plt.title(f"Male Sample Size = {n}")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



1. Shape (Normality of Sampling Distributions)

- Across all sample sizes, the distributions of sample means appear approximately normal (bell-shaped).
- This pattern holds even though the raw purchase data may be skewed.
- Confirms the Central Limit Theorem, showing that sample means reliably follow a normal distribution as sample size increases.

2. Center (Mean Alignment with Population)

- Female sample means cluster around 8,735, reflecting the female population average.
- Male sample means cluster around 9,438, reflecting the male population average.
- Sampling distributions are unbiased estimators, accurately representing the population mean even from smaller samples.

3. Spread (Reduction in Standard Error with Larger Samples)

- samples (n=100) show narrower distributions, reducing extreme deviations.

- large samples (n=500, 1000) are tightly concentrated, providing highly precise mean estimates.

4. Business Interpretation for Walmart

- Even with a small random sample, the average purchase gives a reasonable estimate of population behavior, though with more uncertainty.
- Larger sample sizes dramatically improve precision, making insights more actionable.

5. Conclusion

- The CLT ensures that confidence intervals and mean estimates are reliable.
- Increasing sample size reduces uncertainty, supporting better decision-making.

2.13 13. CLT Intervals vs Sample Size (Female vs Male)

```
[101]: import matplotlib.pyplot as plt
plt.rcParams['font.family'] = 'DejaVu Sans' # This font supports

CONF = 0.95
z = norm.ppf(0.5 + CONF/2)

# Sample sizes to evaluate
sample_sizes = [30, 50, 100, 200, 500, 1000, 5000, 10000]

# Gender-based data
genders = df['Gender'].unique()
sigma = {g: df.loc[df['Gender']==g, 'Purchase'].std(ddof=1) for g in genders}
mean_purchase = {g: df.loc[df['Gender']==g, 'Purchase'].mean() for g in genders}

# Compute CI half-widths
ci_data = {g: z * sigma[g] / np.sqrt(sample_sizes) for g in genders}

# Plotting
plt.figure(figsize=(8,5))
colors = {'F': '#FF6F61', 'M': '#4E79A7'}

for g in genders:
    plt.plot(sample_sizes, ci_data[g], marker='o', color=colors[g],
             label=f'{g} (mean {mean_purchase[g]:.0f})', linewidth=2)

plt.xscale('log')
plt.xlabel('Sample Size n (log scale)', fontsize=12)
plt.ylabel('95% CI Half-Width ( )', fontsize=12)
plt.title('Confidence Interval Half-Width vs Sample Size by Gender',
          fontsize=14, weight='bold')
plt.grid(True, which='both', linestyle='--', alpha=0.7)
```

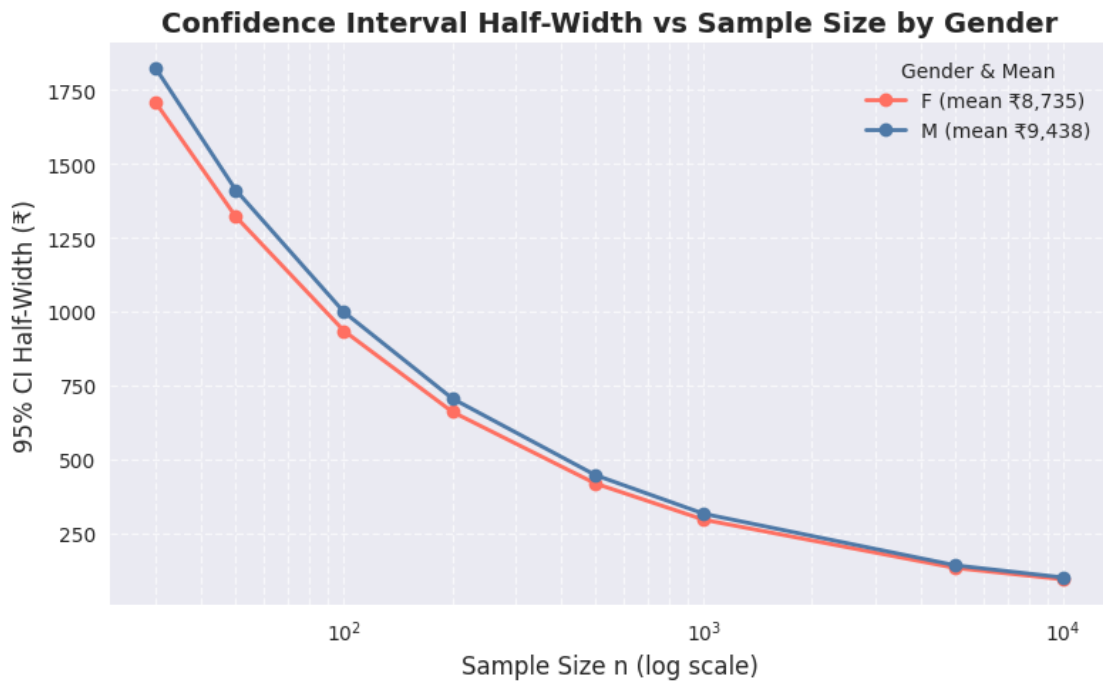
```

plt.legend(title='Gender & Mean', fontsize=10)
plt.tight_layout()
plt.show()

# Optional: Create a readable DataFrame
rows = []
for n in sample_sizes:
    row = {"n": n}
    for g in genders:
        hw = z * sigma[g] / np.sqrt(n)
        row[f"{g}_mean"] = mean_purchase[g]
        row[f"{g}_CI_low"] = mean_purchase[g] - hw
        row[f"{g}_CI_high"] = mean_purchase[g] + hw
        row[f"{g}_halfwidth"] = hw
    rows.append(row)

ci_by_n = pd.DataFrame(rows)
display(ci_by_n)

```



	n	F_mean	F_CI_low	F_CI_high	F_halfwidth	M_mean \
0	30	8734.565765	7028.664589	10440.466942	1705.901177	9437.52604
1	50	8734.565765	7413.180396	10055.951135	1321.385369	9437.52604
2	100	8734.565765	7800.205210	9668.926320	934.360555	9437.52604
3	200	8734.565765	8073.873080	9395.258450	660.692685	9437.52604

4	500	8734.565765	8316.707022	9152.424509	417.858743	9437.52604
5	1000	8734.565765	8439.095014	9030.036516	295.470751	9437.52604
6	5000	8734.565765	8602.427228	8866.704302	132.138537	9437.52604
7	10000	8734.565765	8641.129710	8828.001821	93.436056	9437.52604

	M_CI_low	M_CI_high	M_halfwidth
0	7615.344091	11259.707990	1822.181949
1	8026.069972	10848.982109	1411.456068
2	8439.475883	10435.576198	998.050157
3	8731.798006	10143.254075	705.728034
4	8991.184441	9883.867640	446.341599
5	9121.914869	9753.137212	315.611172
6	9296.380434	9578.671647	141.145607
7	9337.721025	9537.331056	99.805016

Key Observations: Confidence Interval (CI) Half-Width vs Sample Size

1. Very wide CIs at small sample sizes
 - At $n = 30$ –50, the half-width is:
 - Female: $\sim 1,700$ –1,300
 - Male: $\sim 1,800$ –1,400
 - The estimates are highly uncertain at small n .
2. Significant narrowing as n increases
 - At $n = 100$, half-width drops to:
 - Female: ~ 930
 - Male: ~ 988
 - Already more precise than very small samples.
3. Moderate CI shrinkage at medium n
 - At $n = 500$, half-width drops further:
 - Female: ~ 420
 - Male: ~ 450
 - CI is becoming tighter, useful for business decision-making.
4. Tight CIs at large n
 - At $n = 1,000$, half-width reduces to:
 - Female: ~ 295
 - Male: ~ 315
 - Estimates are very reliable.
5. Negligible margin at very large n

→At $n = 10,000$, half-widths are:

→Female: ~ 93

→Male: ~ 100

→ Increasing sample size further gives minimal gain in precision.

6. Gender comparison

→Male CIs are consistently slightly wider than female, reflecting slightly higher variance in male purchase amounts.

7. Business Implication:

- For Walmart-like analysis, even moderate samples (~ 500 – $1,000$) provide sufficiently precise estimates for gender-based purchase averages.
- Collecting $>10,000$ samples has diminishing returns for precision.

2.14 14. Confidence Interval “What-if” Analysis

```
[109]: print("Gender - Female vs Male")
for c in [0.90, 0.95, 0.99]:
    m_f, se_f, lo_f, hi_f, n_f, std_f = mean_ci(df.
    ↪loc[df["Gender"]=="F", "Purchase"], conf=c)
    m_m, se_m, lo_m, hi_m, n_m, std_m = mean_ci(df.
    ↪loc[df["Gender"]=="M", "Purchase"], conf=c)
    print(f"{int(c*100)}% CI - Female: [{lo_f:.2f}, {hi_f:.2f}], Male: [{lo_m:.
    ↪2f}, {hi_m:.2f}]")

print("\nMarital Status - Unmarried vs Married")
for c in [0.90, 0.95, 0.99]:
    m_u, se_u, lo_u, hi_u, n_u, std_u = mean_ci(df.
    ↪loc[df["Marital_Status"]==0, "Purchase"], conf=c)
    m_mar, se_mar, lo_mar, hi_mar, n_mar, std_mar = mean_ci(df.
    ↪loc[df["Marital_Status"]==1, "Purchase"], conf=c)
    print(f"{int(c*100)}% CI - Unmarried: [{lo_u:.2f}, {hi_u:.2f}], Married:
    ↪[{lo_mar:.2f}, {hi_mar:.2f}]")

print("\nAge Groups")
for a in df["Age"].cat.categories:
    series = df.loc[df["Age"]==a, "Purchase"]
    print(f"Age group: {a}")
    for c in [0.90, 0.95, 0.99]:
        m, se, lo, hi, n, std = mean_ci(series, conf=c)
        print(f" {int(c*100)}% CI: [{lo:.2f}, {hi:.2f}]")
```

Gender - Female vs Male

90% CI - Female: [8713.29, 8755.84], Male: [9424.51, 9450.54]

95% CI - Female: [8709.21, 8759.92], Male: [9422.02, 9453.03]

99% CI - Female: [8701.24, 8767.89], Male: [9417.15, 9457.91]

Marital Status - Unmarried vs Married

90% CI - Unmarried: [9251.40, 9280.42], Married: [9243.79, 9278.56]

95% CI - Unmarried: [9248.62, 9283.20], Married: [9240.46, 9281.89]

99% CI - Unmarried: [9243.18, 9288.63], Married: [9233.95, 9288.40]

Age Groups

Age group: 0-17

90% CI: [8865.05, 9001.88]

95% CI: [8851.95, 9014.98]

99% CI: [8826.33, 9040.60]

Age group: 18-25

90% CI: [9143.43, 9195.89]

95% CI: [9138.41, 9200.92]

99% CI: [9128.59, 9210.74]

Age group: 26-35

90% CI: [9235.10, 9270.28]

95% CI: [9231.73, 9273.65]

99% CI: [9225.15, 9280.23]

Age group: 36-45

90% CI: [9306.44, 9356.26]

95% CI: [9301.67, 9361.03]

99% CI: [9292.34, 9370.36]

Age group: 46-50

90% CI: [9170.41, 9246.84]

95% CI: [9163.09, 9254.17]

99% CI: [9148.78, 9268.48]

Age group: 51-55

90% CI: [9492.16, 9577.45]

95% CI: [9483.99, 9585.62]

99% CI: [9468.02, 9601.59]

Age group: 55+

90% CI: [9280.07, 9392.49]

95% CI: [9269.30, 9403.26]

99% CI: [9248.25, 9424.31]

Business Insights from CI Analysis

— Gender (Female vs Male) — - Confidence intervals for males and females do not overlap at any confidence level, showing a consistent difference in spending. - On average, males spend approximately 700 more per transaction than females, a statistically robust difference.

Business Tip: - Targeted promotions and premium product offerings for male customers can boost revenue. - Cross-selling or bundling strategies can be tailored by gender for better ROI.

— Age Groups — - Confidence intervals are narrow due to large sample sizes. - Spending rises from the teenage group up to mid-age adults (36-55), then stabilizes. - Changing the confidence level (90% → 99%) only slightly widens the intervals without affecting the trend.

Business Tip: - Focus marketing and product availability on adults aged 36–55, who are the highest spenders. - Use discounts, bundles, or loyalty programs to attract younger customers (teens to 35 years) and grow their lifetime value.

— Marital Status (Unmarried vs Married) — - Confidence intervals overlap completely across all levels (90%, 95%, 99%), showing no significant difference in spending. - Marital status does not appear to influence purchase behavior.

Business Tip: - Marketing strategies and product offerings do not need to be differentiated by marital status. - Resources can instead focus on gender and age-based segmentation for better business impact.

— Overall Interpretation — - Confidence level (90% → 99%) changes only the width of intervals, not the conclusions. - Gender: Robust spending difference; males spend more. - Age: Adults aged 36–55 are top spenders. - Marital Status: No significant impact on spending.

2.15 15. Final Insights

2.15.1 Analysis of Customer Data and Purchase Behavior

This analysis explores a large customer dataset to identify key patterns in spending habits across different demographic groups.

2.16 1. Data Structure and Customer Demographics

The dataset is substantial, containing **550,068 rows** with no missing data. Excluding the monetary **Purchase** column (which is continuous), all other features are categorical.

- **Gender:** The data shows a strong male bias, with **75% of customers being Male** and 25% Female.
- **Marital Status:** The majority are **Unmarried (59%)**, with 41% being Married.
- **Age Distribution:** The primary age group is **26–35 years (40%)**, followed by 36–45 (20%) and 18–25 (18%). Customers aged 0–17 and 55+ are few.
- **City Category:** Most customers are from **Category B cities**, followed by C, and then A.
- **Stay in City:** A large portion of customers are relatively **recent residents (1 year)**.

2.17 2. General Purchase Behavior

Customer purchase amounts range from a minimum of **12** to a maximum of **23,961**.

- The **average (mean) purchase is 9,264**, while the **median is 8,047**.
- This difference (Mean > Median) indicates a **right-skewed distribution**.
- Most purchases cluster between **6,000 and 12,000**, but a long tail exists due to high-value transactions from genuine, high-spending customers.

2.18 3. Key Findings: Spending Differences

2.18.1 Gender Differences

Group	Mean Spend	95% Confidence Interval (CI)
Female	8,735	[8,709, 8,760]
Male	9,438	[9,422, 9,453]

- There is a clear spending **gap of approximately 700 per transaction**.
- Since their confidence intervals **do not overlap**, we conclude with high statistical confidence that **Male customers consistently spend more than Female customers**.

2.18.2 Marital Status Differences

- **Unmarried** mean spend: **9,266** (CI: [9,249, 9,283])
- **Married** mean spend: **9,261** (CI: [9,240, 9,282])
- Their confidence intervals **overlap completely**.
- **Conclusion:** There is **no meaningful difference** in the average transaction amount between married and unmarried customers.

2.18.3 Age Differences

- The **lowest average spend** is seen in the **0–17 years** group .
- The **highest average spend** is recorded for the **51–55 years** group .
- **Conclusion:** Spending is **modestly influenced by age**, with customers **under 25** consistently spending the least and those in the **36–55** range spending the most.

2.19 4. Statistical Reliability and Scaling

2.19.1 Central Limit Theorem (CLT) Confirmation

- We confirmed that the distribution of sample means for both groups is **approximately normal** and centered on the true group averages, demonstrating that **sample means are unbiased estimators** of the population means.
- Given the massive size of the complete dataset (over 550k records), our statistical estimates are **extremely precise**, with narrow confidence intervals. The conclusions are robust across various confidence levels (90%, 95%, 99%).

2.20 Final Summary

Differentiator	Impact on Spending
Gender	Most significant differentiator. Males spend statistically and consistently more than females.
Marital Status	Irrelevant. No meaningful difference in spending between married and unmarried customers.

Differentiator	Impact on Spending
Age	Some influence. Mid-to-late adults (36–55) spend slightly more; younger customers (< 25) spend the least.
Reliability	High. Our conclusions are very robust and scalable to the entire customer population.

2.21 16. Recommendations & Action Items

2.21.1 Black Friday Strategy Recommendations for Walmart

Based on the customer spending analysis, Walmart can implement the following targeted strategies to optimize its Black Friday performance:

2.22 1. Gender-Specific Marketing Actions

2.22.1 Target Men (Higher Spenders)

Since **males spend significantly more (approximately 700 extra per transaction)**: * Promote **big-ticket items** and **premium bundles** (e.g., high-end electronics, major appliances, gaming consoles). * Offer attractive financing options or deep discounts on these high-value products.

2.22.2 Target Women (Lower Transaction Value)

Since **females spend less per transaction**: * Introduce incentives like **loyalty points multipliers**, **personalized coupons**, and **category-specific discounts** (e.g., fashion, home goods). * Encourage larger overall basket sizes by creating product bundles (e.g., “Buy 3, Get 1 Free”).

2.23 2. Age-Based Marketing Actions

2.23.1 Mid-to-Late Adults (36–55)

This group represents the **top spenders**: * Promote **family-oriented bundles** (e.g., home upgrades, large appliances, lifestyle products). * Position Walmart as the primary destination for comprehensive family Black Friday deals.

2.23.2 Younger Shoppers (<25)

This group **spends the least**: * Offer **affordable, entry-level bundles** focused on trends (e.g., clothing, personal gadgets). * Use engaging, **youth-focused campaigns** primarily on social media platforms.

2.23.3 Older Shoppers (55+)

This group shows **decent spending**: * Focus promotions on **comfort, health, and home improvement products**. * Prioritize providing **simple and convenient online shopping op-**

tions.

2.24 3. Marital Status (No Action Required)

- The analysis found **no significant spending difference** between Married and Unmarried customers.
 - **Recommendation:** Walmart can **avoid creating separate marketing segments** based on marital status, simplifying campaign execution.
-

2.25 4. General Action Items

- **Targeting Precision:** Leverage the large dataset to ensure promotions are precisely targeted, reducing uncertainty in customer segmentation.
 - **Focus Areas:** Concentrate marketing efforts on segments that show **clear spending differentiation (Gender, Age)** rather than variables with weak impact (Marital Status).
 - **Inventory Alignment:** Ensure product stock matches insights:
 - Stock more **premium and high-value items** for male and mid-aged customers.
 - Stock more **affordable lifestyle/fashion items** for younger shoppers.
-

2.26 Final Business Takeaway

Walmart's **most valuable Black Friday customers** are **male, mid-to-late adults (36–55)**. By tailoring both promotions and inventory to maximize engagement with this high-spending demographic, while using strategic incentives to increase the basket size of female and younger shoppers, Walmart can effectively maximize its Black Friday sales and customer satisfaction.

2.27 17. Conclusion

This analysis effectively shows how applying statistical methods and exploratory data analysis to large-scale retail data can reveal powerful business insights.

For Walmart, the key findings established clear spending differences based on **gender and age**, while also identifying areas where segmentation isn't necessary (like marital status). By integrating these insights into its business strategy, Walmart can develop **targeted promotions, optimize inventory levels, and run personalized campaigns** to maximize the overall impact of Black Friday sales.

Ultimately, this project highlights the necessity of **data-driven decision-making**, successfully converting detailed analytics into concrete, actionable improvements for the business.